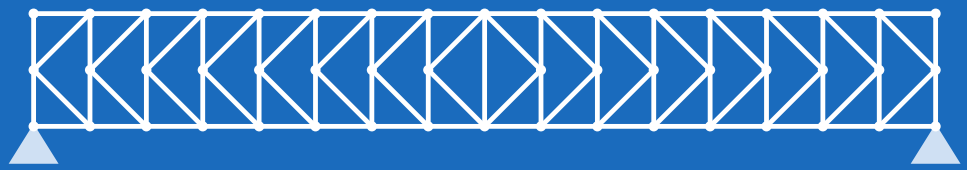
TRUSS REPORT

K-truss bridge (97 members)

Plane truss — analysis and CIRSOC 301 rod checks



Author A. Student · Course Structures I · 13/10/2026

MAX TENSION	MAX COMPRESSION	DEFLECTION	MOST USED ROD	BUCKLING Λ_1	WEIGHT
+945.0 kN	-945.0 kN	84.1 mm	0.73	2.34	15679 kg
rod 8	rod 24	L/761 · limit L/300	rod 8 · normal stress (H.3.4)	the load can grow ×2.3	carries 900.0 kN · 6× own weight

The model

Span	64.00 m
Height	8.00 m
Nodes / rods	50 / 97
Supports	pin at 0, rollerX at 16
Loads	15 nodal, total 900.0 kN
Rod families	5, with their sections:
Bottom chord	HEA 220, F24 (CIRSOC) / A36
Diagonal	SHS 150x150x6, F24 (CIRSOC) / A36
End post	SHS 200x200x8, F24 (CIRSOC) / A36
Top chord	HEB 220, F24 (CIRSOC) / A36
Vertical	SHS 120x120x5, F24 (CIRSOC) / A36

What the analysis found

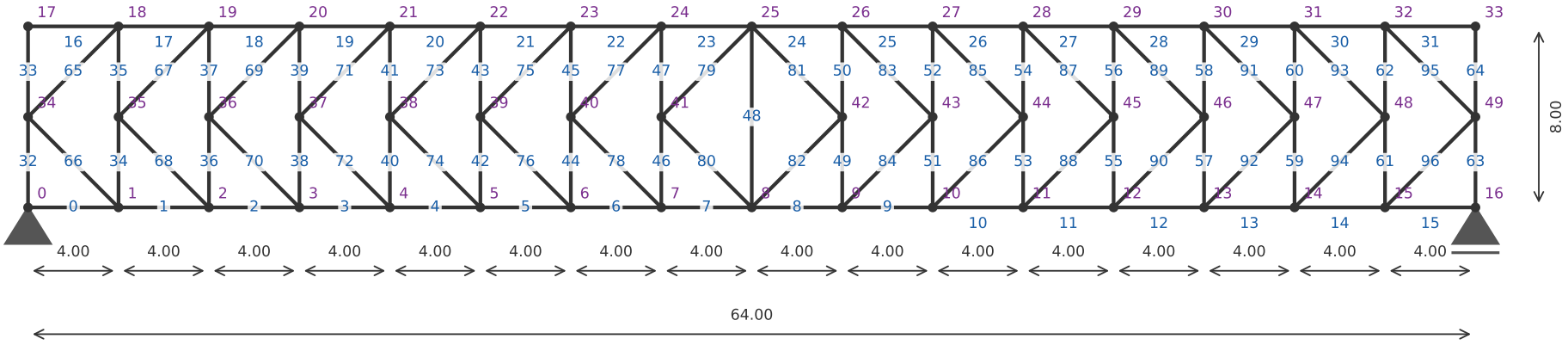
Stability	Statically determinate (m + r − 2j = 97 + 3 − 2×50 = 0): statics alone gives every force.
Most used rod	rod 8 at 0.73 (normal stress (H.3.4))
Rods over capacity	0 of 97 checked
Buckling	$\lambda_1 = 2.34$
Shear and moment	none: every rod is pinned and loaded only at its nodes, so the axial force is the whole result

Contents

2 Geometry
3 Axial forces
4 Deflection and stress
5 Reactions
6 Utilisation
7 Buckling
8 Rods · 2 sheets
10 Explain rod 8 and 7

Results are an aid to a qualified professional, who must check them. Rod checks to CIRSOC 301 as implemented; first-order analysis; joints and connections are not checked here.

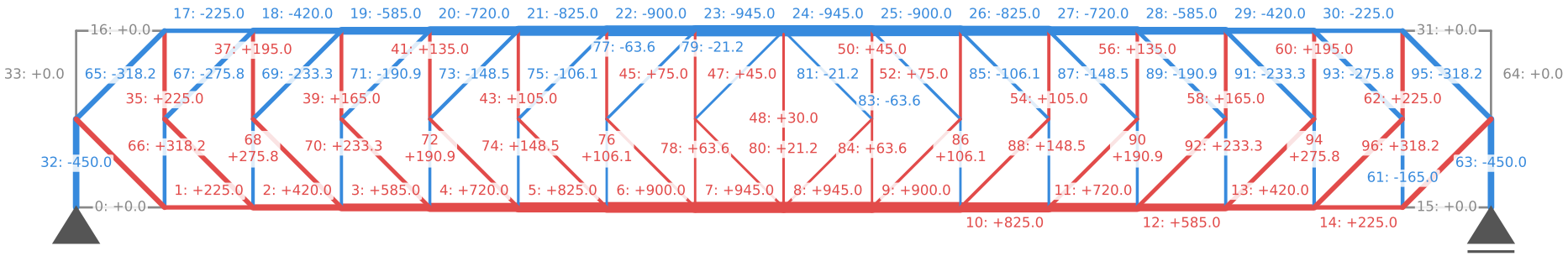
Node numbers in purple, rod numbers in blue. Dimensions in m.



20 m · true to scale

0 20

Axial force N (kN): red pulls (+), blue pushes (−), grey carries nothing; line width follows |N| (largest 945.0).

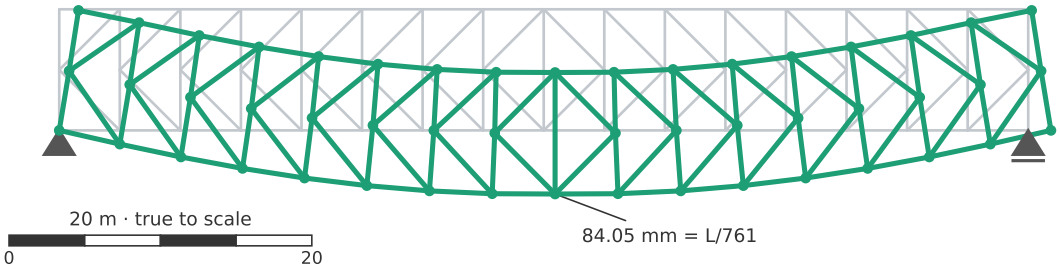


20 m · true to scale

0 20

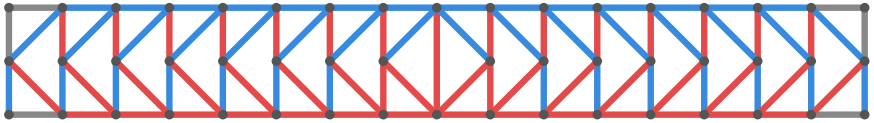
14 of 97 labels left out where they would overlap; every force is in the Rods table.

Elastic line: the deflected shape (green) over the truss (grey), every rigid rod drawn as the curve it bends into. Geometry to scale; displacements magnified × 50. Every rod is pinned, so every member stays straight: there is no curvature, and no contraflexure. Largest vertical deflection 84.05 mm against the limit L/300 = 213.3 mm: OK.

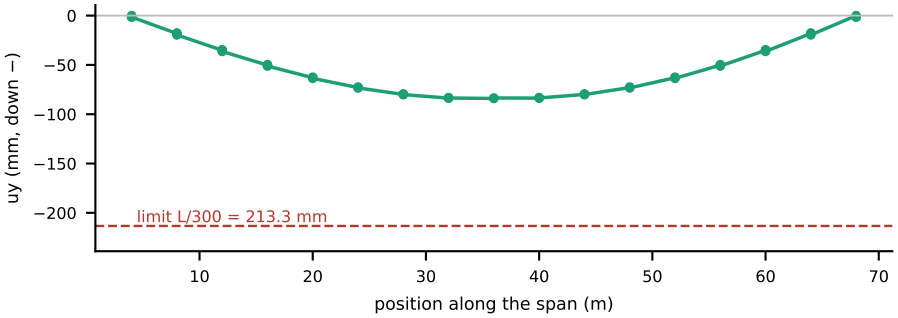


Fibre stress top and bottom face of each member

red = tension, blue = compression. Every rod is pinned, so both faces of a rod carry the same stress and each rod is one colour: the axial force.

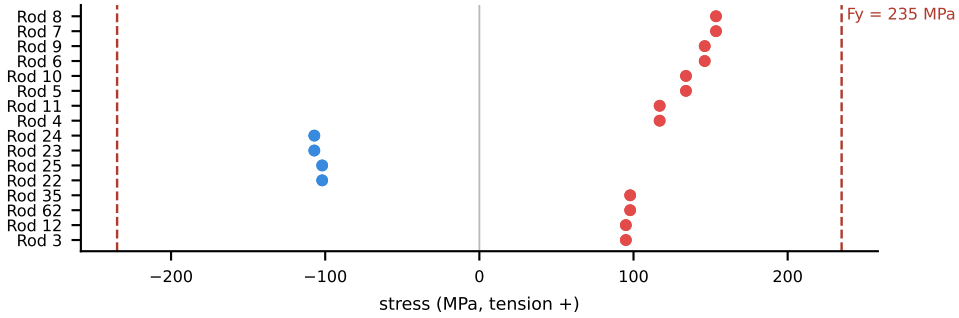


Deflection along the span vertical movement of every joint



Largest 84.05 mm at joint 8 = L/761. Within the limit.

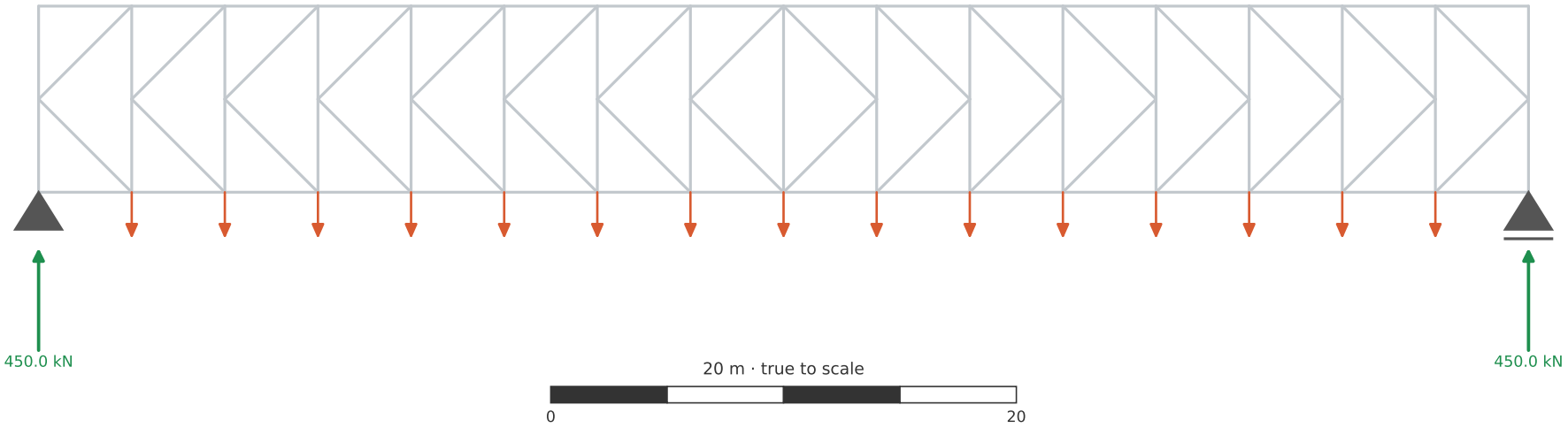
Fibre stress values the two faces of each rod, MPa



Every rod is pinned, so its two fibres carry the same N/A and coincide: no bending opens the pair.

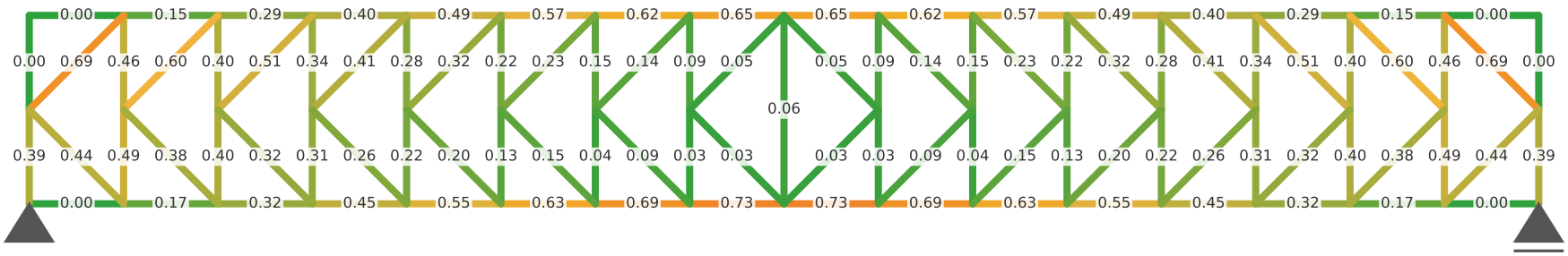
Support reactions (green arrows), as the supports push on the truss, against the loads (orange); Ry positive upwards. The sums balance the loads.

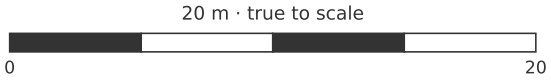
15 loads of 60.0 kN each, hung from the joints

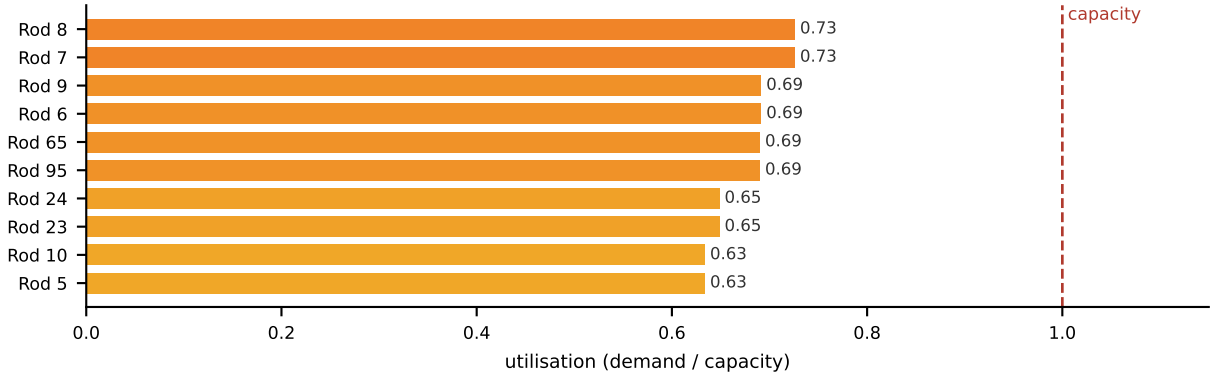


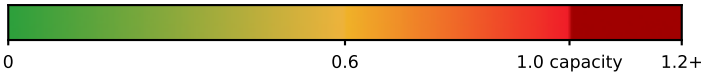
Node	Type	Rx (kN)	Ry (kN)
0	pin	0.00	450.00
16	rollerX	0.00	450.00
Σ		0.00	900.00

Utilisation: demand over CIRSOC 301 capacity. Most used rod 8 at 0.73 (normal stress (H.3.4)). Every checked rod is within capacity.







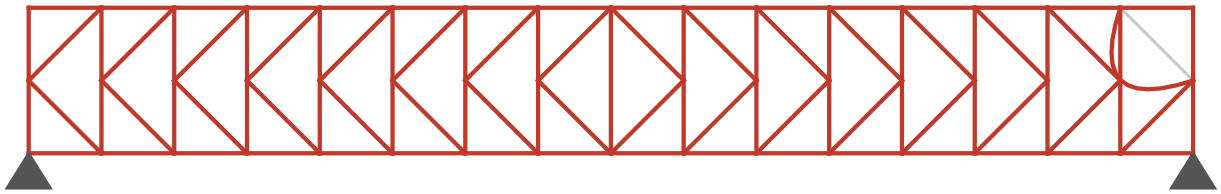


grey = no steel section, not checked

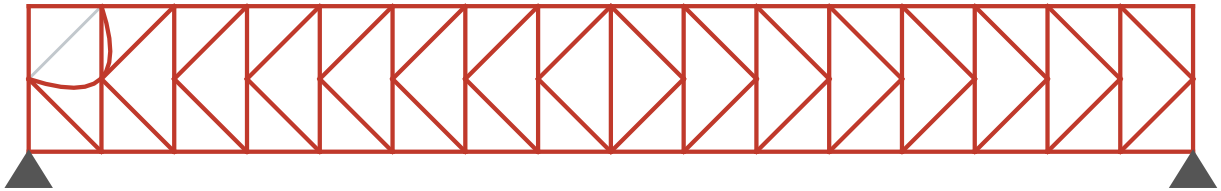
The 10 most used of 97 checked rods; the rest are in the Rods table.

λ is the factor on the load as it stands. $\lambda_1 = 2.34$, so the load can grow 2.3× before it buckles. 2 modes share that λ: that many equal members buckle at the same load, so stiffening one of them buys nothing.

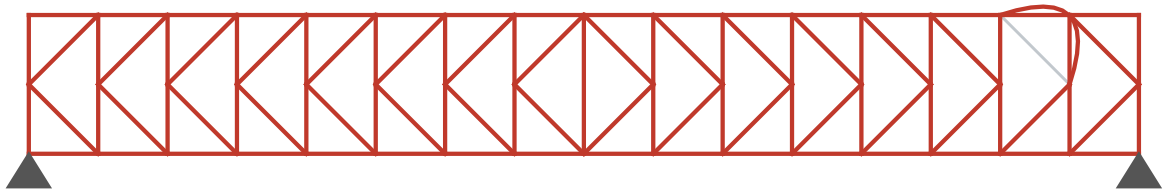
Mode 1: λ = 2.34
rod 95 bows

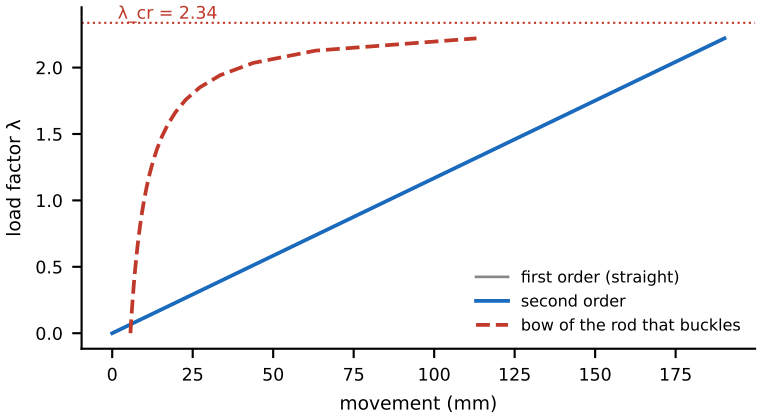


Mode 2: λ = 2.34
rod 65 bows



Mode 3: λ = 2.70
rod 93 bows





The second-order line bends away from the first-order one and runs off as λ nears λ_cr: that gap IS the amplification every second-order design rule is about. Started from an initial bow of L/1000 = 5.66 mm in the first mode's shape.

Elastic buckling of the IDEAL truss, so every λ is an UPPER bound: imperfections, residual stresses and yielding are not modelled, and each rod is cut in two so it can bow between its joints. Buckling OUT of the plane is the rod check's job (CIRSOC 301 E3, minor radius) and is in the Utilisation and Rods sheets, not here. A mode has a shape, not a size: the shapes are drawn exaggerated.

Every rod: its end nodes, family and steel section, length, axial force N (+ pulls, − pushes), the CIRSOC 301 check that governs it and how much of its capacity that uses (Use = demand / capacity; OVER above 1.0).

Rod	Nodes	Family	Section	L (m)	N (kN)	Governing check	Use
0	0–1	Bottom chord	HEA 220	4.00	+0.00	inelastic buckling (E3-2)	0.00 OK
1	1–2	Bottom chord	HEA 220	4.00	+225.00	normal stress (H.3.4)	0.17 OK
2	2–3	Bottom chord	HEA 220	4.00	+420.00	normal stress (H.3.4)	0.32 OK
3	3–4	Bottom chord	HEA 220	4.00	+585.00	normal stress (H.3.4)	0.45 OK
4	4–5	Bottom chord	HEA 220	4.00	+720.00	normal stress (H.3.4)	0.55 OK
5	5–6	Bottom chord	HEA 220	4.00	+825.00	normal stress (H.3.4)	0.63 OK
6	6–7	Bottom chord	HEA 220	4.00	+900.00	normal stress (H.3.4)	0.69 OK
7	7–8	Bottom chord	HEA 220	4.00	+945.00	normal stress (H.3.4)	0.73 OK
8	8–9	Bottom chord	HEA 220	4.00	+945.00	normal stress (H.3.4)	0.73 OK
9	9–10	Bottom chord	HEA 220	4.00	+900.00	normal stress (H.3.4)	0.69 OK
10	10–11	Bottom chord	HEA 220	4.00	+825.00	normal stress (H.3.4)	0.63 OK
11	11–12	Bottom chord	HEA 220	4.00	+720.00	normal stress (H.3.4)	0.55 OK
12	12–13	Bottom chord	HEA 220	4.00	+585.00	normal stress (H.3.4)	0.45 OK
13	13–14	Bottom chord	HEA 220	4.00	+420.00	normal stress (H.3.4)	0.32 OK
14	14–15	Bottom chord	HEA 220	4.00	+225.00	normal stress (H.3.4)	0.17 OK
15	15–16	Bottom chord	HEA 220	4.00	+0.00	inelastic buckling (E3-2)	0.00 OK
16	17–18	Top chord	HEB 220	4.00	+0.00	normal stress (H.3.4)	0.00 OK
17	18–19	Top chord	HEB 220	4.00	-225.00	inelastic buckling (E3-2)	0.15 OK
18	19–20	Top chord	HEB 220	4.00	-420.00	inelastic buckling (E3-2)	0.29 OK
19	20–21	Top chord	HEB 220	4.00	-585.00	inelastic buckling (E3-2)	0.40 OK
20	21–22	Top chord	HEB 220	4.00	-720.00	inelastic buckling (E3-2)	0.49 OK
21	22–23	Top chord	HEB 220	4.00	-825.00	inelastic buckling (E3-2)	0.57 OK
22	23–24	Top chord	HEB 220	4.00	-900.00	inelastic buckling (E3-2)	0.62 OK
23	24–25	Top chord	HEB 220	4.00	-945.00	inelastic buckling (E3-2)	0.65 OK
24	25–26	Top chord	HEB 220	4.00	-945.00	inelastic buckling (E3-2)	0.65 OK
25	26–27	Top chord	HEB 220	4.00	-900.00	inelastic buckling (E3-2)	0.62 OK
26	27–28	Top chord	HEB 220	4.00	-825.00	inelastic buckling (E3-2)	0.57 OK
27	28–29	Top chord	HEB 220	4.00	-720.00	inelastic buckling (E3-2)	0.49 OK
28	29–30	Top chord	HEB 220	4.00	-585.00	inelastic buckling (E3-2)	0.40 OK
29	30–31	Top chord	HEB 220	4.00	-420.00	inelastic buckling (E3-2)	0.29 OK
30	31–32	Top chord	HEB 220	4.00	-225.00	inelastic buckling (E3-2)	0.15 OK
31	32–33	Top chord	HEB 220	4.00	+0.00	inelastic buckling (E3-2)	0.00 OK
32	0–34	End post	SHS 200x200x8	4.00	-450.00	inelastic buckling (E3-2)	0.39 OK
33	34–17	End post	SHS 200x200x8	4.00	+0.00	normal stress (H.3.4)	0.00 OK
34	1–35	Vertical	SHS 120x120x5	4.00	-165.00	inelastic buckling (E3-2)	0.49 OK
35	35–18	Vertical	SHS 120x120x5	4.00	+225.00	normal stress (H.3.4)	0.46 OK
36	2–36	Vertical	SHS 120x120x5	4.00	-135.00	inelastic buckling (E3-2)	0.40 OK
37	36–19	Vertical	SHS 120x120x5	4.00	+195.00	normal stress (H.3.4)	0.40 OK
38	3–37	Vertical	SHS 120x120x5	4.00	-105.00	inelastic buckling (E3-2)	0.31 OK
39	37–20	Vertical	SHS 120x120x5	4.00	+165.00	normal stress (H.3.4)	0.34 OK
40	4–38	Vertical	SHS 120x120x5	4.00	-75.00	inelastic buckling (E3-2)	0.22 OK
41	38–21	Vertical	SHS 120x120x5	4.00	+135.00	normal stress (H.3.4)	0.28 OK
42	5–39	Vertical	SHS 120x120x5	4.00	-45.00	inelastic buckling (E3-2)	0.13 OK
43	39–22	Vertical	SHS 120x120x5	4.00	+105.00	normal stress (H.3.4)	0.22 OK
44	6–40	Vertical	SHS 120x120x5	4.00	-15.00	inelastic buckling (E3-2)	0.04 OK
45	40–23	Vertical	SHS 120x120x5	4.00	+75.00	normal stress (H.3.4)	0.15 OK
46	7–41	Vertical	SHS 120x120x5	4.00	+15.00	normal stress (H.3.4)	0.03 OK
47	41–24	Vertical	SHS 120x120x5	4.00	+45.00	normal stress (H.3.4)	0.09 OK
48	8–25	Vertical	SHS 120x120x5	8.00	+30.00	normal stress (H.3.4)	0.06 OK
49	9–42	Vertical	SHS 120x120x5	4.00	+15.00	normal stress (H.3.4)	0.03 OK
50	42–26	Vertical	SHS 120x120x5	4.00	+45.00	normal stress (H.3.4)	0.09 OK
51	10–43	Vertical	SHS 120x120x5	4.00	-15.00	inelastic buckling (E3-2)	0.04 OK

Every rod: its end nodes, family and steel section, length, axial force N (+ pulls, − pushes), the CIRSOC 301 check that governs it and how much of its capacity that uses (Use = demand / capacity; OVER above 1.0).

Rod	Nodes	Family	Section	L (m)	N (kN)	Governing check	Use
52	43–27	Vertical	SHS 120x120x5	4.00	+75.00	normal stress (H.3.4)	0.15 OK
53	11–44	Vertical	SHS 120x120x5	4.00	-45.00	inelastic buckling (E3-2)	0.13 OK
54	44–28	Vertical	SHS 120x120x5	4.00	+105.00	normal stress (H.3.4)	0.22 OK
55	12–45	Vertical	SHS 120x120x5	4.00	-75.00	inelastic buckling (E3-2)	0.22 OK
56	45–29	Vertical	SHS 120x120x5	4.00	+135.00	normal stress (H.3.4)	0.28 OK
57	13–46	Vertical	SHS 120x120x5	4.00	-105.00	inelastic buckling (E3-2)	0.31 OK
58	46–30	Vertical	SHS 120x120x5	4.00	+165.00	normal stress (H.3.4)	0.34 OK
59	14–47	Vertical	SHS 120x120x5	4.00	-135.00	inelastic buckling (E3-2)	0.40 OK
60	47–31	Vertical	SHS 120x120x5	4.00	+195.00	normal stress (H.3.4)	0.40 OK
61	15–48	Vertical	SHS 120x120x5	4.00	-165.00	inelastic buckling (E3-2)	0.49 OK
62	48–32	Vertical	SHS 120x120x5	4.00	+225.00	normal stress (H.3.4)	0.46 OK
63	16–49	End post	SHS 200x200x8	4.00	-450.00	inelastic buckling (E3-2)	0.39 OK
64	49–33	End post	SHS 200x200x8	4.00	+0.00	normal stress (H.3.4)	0.00 OK
65	34–18	Diagonal	SHS 150x150x6	5.66	-318.20	inelastic buckling (E3-2)	0.69 OK
66	34–1	Diagonal	SHS 150x150x6	5.66	+318.20	normal stress (H.3.4)	0.44 OK
67	35–19	Diagonal	SHS 150x150x6	5.66	-275.77	inelastic buckling (E3-2)	0.60 OK
68	35–2	Diagonal	SHS 150x150x6	5.66	+275.77	normal stress (H.3.4)	0.38 OK
69	36–20	Diagonal	SHS 150x150x6	5.66	-233.35	inelastic buckling (E3-2)	0.51 OK
70	36–3	Diagonal	SHS 150x150x6	5.66	+233.35	normal stress (H.3.4)	0.32 OK
71	37–21	Diagonal	SHS 150x150x6	5.66	-190.92	inelastic buckling (E3-2)	0.41 OK
72	37–4	Diagonal	SHS 150x150x6	5.66	+190.92	normal stress (H.3.4)	0.26 OK
73	38–22	Diagonal	SHS 150x150x6	5.66	-148.49	inelastic buckling (E3-2)	0.32 OK
74	38–5	Diagonal	SHS 150x150x6	5.66	+148.49	normal stress (H.3.4)	0.20 OK
75	39–23	Diagonal	SHS 150x150x6	5.66	-106.07	inelastic buckling (E3-2)	0.23 OK
76	39–6	Diagonal	SHS 150x150x6	5.66	+106.07	normal stress (H.3.4)	0.15 OK
77	40–24	Diagonal	SHS 150x150x6	5.66	-63.64	inelastic buckling (E3-2)	0.14 OK
78	40–7	Diagonal	SHS 150x150x6	5.66	+63.64	normal stress (H.3.4)	0.09 OK
79	41–25	Diagonal	SHS 150x150x6	5.66	-21.21	inelastic buckling (E3-2)	0.05 OK
80	41–8	Diagonal	SHS 150x150x6	5.66	+21.21	normal stress (H.3.4)	0.03 OK
81	42–25	Diagonal	SHS 150x150x6	5.66	-21.21	inelastic buckling (E3-2)	0.05 OK
82	42–8	Diagonal	SHS 150x150x6	5.66	+21.21	normal stress (H.3.4)	0.03 OK
83	43–26	Diagonal	SHS 150x150x6	5.66	-63.64	inelastic buckling (E3-2)	0.14 OK
84	43–9	Diagonal	SHS 150x150x6	5.66	+63.64	normal stress (H.3.4)	0.09 OK
85	44–27	Diagonal	SHS 150x150x6	5.66	-106.07	inelastic buckling (E3-2)	0.23 OK
86	44–10	Diagonal	SHS 150x150x6	5.66	+106.07	normal stress (H.3.4)	0.15 OK
87	45–28	Diagonal	SHS 150x150x6	5.66	-148.49	inelastic buckling (E3-2)	0.32 OK
88	45–11	Diagonal	SHS 150x150x6	5.66	+148.49	normal stress (H.3.4)	0.20 OK
89	46–29	Diagonal	SHS 150x150x6	5.66	-190.92	inelastic buckling (E3-2)	0.41 OK
90	46–12	Diagonal	SHS 150x150x6	5.66	+190.92	normal stress (H.3.4)	0.26 OK
91	47–30	Diagonal	SHS 150x150x6	5.66	-233.35	inelastic buckling (E3-2)	0.51 OK
92	47–13	Diagonal	SHS 150x150x6	5.66	+233.35	normal stress (H.3.4)	0.32 OK
93	48–31	Diagonal	SHS 150x150x6	5.66	-275.77	inelastic buckling (E3-2)	0.60 OK
94	48–14	Diagonal	SHS 150x150x6	5.66	+275.77	normal stress (H.3.4)	0.38 OK
95	49–32	Diagonal	SHS 150x150x6	5.66	-318.20	inelastic buckling (E3-2)	0.69 OK
96	49–15	Diagonal	SHS 150x150x6	5.66	+318.20	normal stress (H.3.4)	0.44 OK



Rod 8

utilisation 0.73

normal stress (H.3.4)

```
Rod 8
=====
Steel, CIRSOC 301-2018 -- A = 6,156 mm², Fy = 235 MPa, L = 4.000 m

 1. Axial stress
      σ = N / A
      = 945,000 N / 6,156 mm²
      = 153.51 MPa

 2. Design stress  [H.3.4]
      φ · Fy
      = 0.9 × 235
      = 211.50 MPa

 3. Tension ratio  [H.3.4]
      σ / (φ Fy)
      = 153.51 / 211.50
      = 0.726

 4. Governs
      the largest ratio
      = normal stress (H.3.4)
      = 0.726

utilisation 0.726 -- OK
```



Rod 7

utilisation 0.73

normal stress (H.3.4)

```
Rod 7
=====
Steel, CIRSOC 301-2018 -- A = 6,156 mm², Fy = 235 MPa, L = 4.000 m

 1. Axial stress
      σ = N / A
      = 945,000 N / 6,156 mm²
      = 153.51 MPa

 2. Design stress  [H.3.4]
      φ · Fy
      = 0.9 × 235
      = 211.50 MPa

 3. Tension ratio  [H.3.4]
      σ / (φ Fy)
      = 153.51 / 211.50
      = 0.726

 4. Governs
      the largest ratio
      = normal stress (H.3.4)
      = 0.726

utilisation 0.726 -- OK
```